

Omer Bin Habib

Senior Software Engineer (Independent Consultant) | Python · Node.js · PostgreSQL · Redis · CI/CD

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GitHub: <https://github.com/omerhabib62>

PROFESSIONAL SUMMARY

Backend engineer with 6 years' experience building scalable data pipelines, real-time event processing, and analytics infrastructure across B2B SaaS products. MSBA candidate at KSBL (expected 2027), applying financial modelling and decision-science frameworks to product and requirements decisions. Practical experience with **ETL workflows**, **schema design**, **high-throughput webhook** ingestion via **BullMQ/Redis**, and structured-data extraction pipelines. Strong technical foundation in **Python**, **Node.js**, **TypeScript**, **PostgreSQL**, **BullMQ/Redis** event streaming, **CI/CD pipelines**, and **AWS**.

EDUCATION

Master's in Business Analytics (MSBA) | KSBL | Karachi, Pakistan | **Expected: 2027**

Bachelor of Science in Computer Science (BSCS) | SZABIST | Karachi, Pakistan | **Graduated: 2020**

PROFESSIONAL EXPERIENCE

Senior Engineer & Consultant (Independent Consultant) | UK (Remote) Jan 2026 – Present

- Sole backend engineer reporting directly to Founder, responsible for architecture, implementation, testing, and deployment across 3 B2B SaaS products with deep data-pipeline work across each.
- **Spot — Proactive Employee Mental Health Platform**
 - Built the complete NestJS backend (6 modules, 18+ migrations, 357 tests) for an AI-powered mental health check-in platform with schema design, event ingestion, and ETL into structured risk signals.
 - Engineered the clinical risk-scoring pipeline: NLP feature extraction from free-text input, EWMA baseline comparison per user, confidence-gated alert classification, and crisis phrase detection. Designed tiered LLM model routing (Haiku to Sonnet fallback) optimising cost, latency, and accuracy.
 - Built the email verification and bounce-handling pipeline integrating NeverBounce API and Postmark webhooks, with hourly timezone-aware batch jobs and a strict gate preventing unverified contacts from receiving messages.
 - Implemented Stripe seat-based tiered billing with automated proration, debounced sync, and dunning/suspension workflows.
 - Built a GDPR compliance layer covering cross-module data export, anonymization, and hard delete via key-authenticated internal API boundary.
 - Engineered a CSV import pipeline supporting 20+ header variants with three-tier validation (discard, billable error, valid), deduplication, and two-step confirm flow.
- **Groups — Global Community Monetization Platform**
 - Architected backend infrastructure for a tiered creator/community platform (13 modules, 73 migrations, 81 entities) using NestJS Clean Architecture with domain/application/infrastructure separation.
 - Designed financial ledger with Stripe Connect: ACID transactions for tiered commission distribution (TTS) with zero financial discrepancy.
 - Built observability pipeline integrating Pino (structured JSON logging with correlation IDs), Sentry global error tracking, and @nestjs/terminus automated health checks.
 - Engineered API circuit breakers and global rate limiting (100 req/min) for third-party integrations with graceful degradation and zero-downtime deployments.
 - Designed persistent retrievable job queues using BullMQ and Redis (Upstash) for high-throughput webhook ingestion and billing state synchronization.
 - Engineered a hybrid search engine using PostgreSQL pgvector and Redis with cursor-based pagination, reducing database load by 90%.
- **Audit-this-engine — Agency Code Intelligence Platform (B2B SaaS, pre-launch)**
 - Conducted a 6-layer industry and market analysis using MSBA financial modeling frameworks, covering TAM/SAM/SOM sizing, competitor benchmarking, customer segmentation, pricing research, and CAC/sales funnel modeling.
 - Analyzed competitive positioning across key dimensions and recommended discontinuing one product direction in a saturated market (against an \$88M Series B-funded category leader), shifting toward a more defensible alternative.
 - Designed pricing and validation strategy, defining tiers (£149 BYO, £249 managed), validating ROI across regions and segments, and building an 8-question discovery framework for willingness-to-pay testing.
 - Modeled sales funnel economics across 3 scenarios (6-week optimistic to 8-month conservative path to traction), introducing milestone-based checkpoints (week 4 and week 8) instead of fixed commitments to better reflect execution risk.
 - Shipped engine foundation — 7 static-analysis tools orchestrated via per-tool BullMQ fan-out/fan-in decomposition, LLM enrichment pipeline with tool-calling and prompt-caching, scan-from-URL endpoint, diagnostic tool-run tracing.
- **Working Tech stack:** NestJS, TypeORM, PostgreSQL, Redis, Stripe, Postmark, Claude API, OpenAI API, BullMQ, Jest, Sentry, Docker

Software Engineer | Blocship | Karachi, Pakistan

May 2023 – Dec 2025

- Defined SLOs and optimized a real-time commission engine for 10K+ users under strict cost and performance constraints.
- Built requirements for a low-latency messaging system (Socket.io + Redis) with sub-millisecond sync targets.
- Led migration to NestJS/TypeScript and standardized engineering practices, reducing production bugs by 40%.
- Drove API-first adoption, enabling parallel development and faster sprint cycles.

Software Engineer | Sybrid (Pvt) Ltd | Karachi, Pakistan

April 2021 – May 2023

- **FiTE – multi-tenant, multilingual recruitment ecosystem (Japan–Pakistan)**
 - Led requirements definition for a cross-border recruitment platform by conducting stakeholder interviews across 4 personas (candidates, agencies, clients, admins), defining end-to-end vetting workflows and orchestration logic aligned with OWASP security practices.
 - Translated persona requirements into a role-based access system, writing user stories for 4 distinct workflows with granular permissions, profiling, and vetting pipelines.
- **Internal Project Administration System:** Owned internal backlog prioritization, enabling automation of resource allocation across PM teams.

Associate Developer | TAFSOL Technologies (Pvt) Ltd | Karachi, Pakistan

Sept 2020 – April 2021

PROJECTS

Mercenary (SaaS Boilerplate) | NestJS, PostgreSQL, Redis, Stripe:

- Developed a production-ready backend architecture for FinTech and SaaS scalability, implementing RBAC, Stripe subscriptions, Redis-based job queues, and API rate limiting.

Apna Khata (Wholesale Agri-Marketplace) | Laravel, NodeJS, MySQL

- Digitized Lahore wholesale produce market by building a farmer-to-distributor B2B platform that has a real-time auction system automating bidding and price negotiation from offer to settlement. Also, it managed end-to-end trade flow for perishable goods, including inventory ingestion and order fulfillment.

SKILLS, COMPETENCIES & CERTIFICATIONS

Skills:

- **Analytics & Data Tools:** SQL (PostgreSQL daily), Python, pgvector for hybrid search and RAG, Excel/Sheets for analysis, Jira, Notion
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- **Technical Stack:** TypeScript, Python, Node.js, NestJS, NextJS, Laravel, TypeORM, AWS (EC2, RDS, S3), Docker, GitHub Actions, Sentry, Prometheus/Grafana, REST APIs
- **Databases:** PostgreSQL, Redis, SQLite, pgvector
- **AI/LLM:** Claude/OpenAI APIs, LangGraph, DSPy, RAG pipelines, Langfuse
- **Architecture:** Event-Driven Architecture, REST APIs, Agentic SDLC, Distributed Systems

Certifications:

- **Introduction to Cloud Computing (Credential ID: # EL9NUHQT2FZF) IBM, 2025**
<https://www.coursera.org/account/accomplishments/verify/EL9NUHQT2FZF>
- **Introduction to Software Engineering (Credential ID: # JHB51DAZGCT6) IBM, 2025**
<https://www.coursera.org/account/accomplishments/verify/JHB51DAZGCT6>
- **Building RESTful APIs in Laravel (Credential ID: # 6c129a77255a25e7f36fc6d61cc70667b21ebdf23eebe42a29ce65d769d969a4) LinkedIn, 2022**
https://www.linkedin.com/learning/certificates/6c129a77255a25e7f36fc6d61cc70667b21ebdf23eebe42a29ce65d769d969a4?trk=backfilled_certificate
- **PHP: Creating Secure Websites (Credential ID: 90fe2123766b7f47bb6fb43ce1de912b3515040f7f748155e3373c587b7cc07c) LinkedIn, 2022**
https://www.linkedin.com/learning/certificates/90fe2123766b7f47bb6fb43ce1de912b3515040f7f748155e3373c587b7cc07c?trk=backfilled_certificate